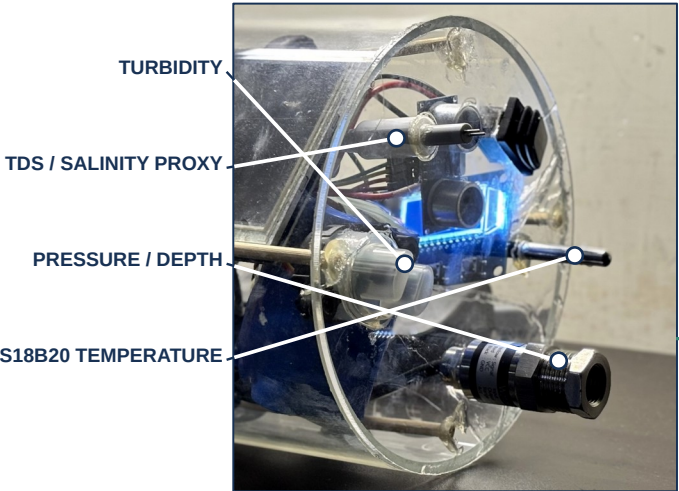


SENSE · REAL ENVIRONMENT

Live water-condition inputs



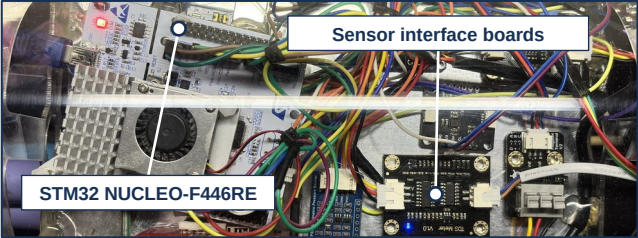
ADC · 1-Wire · I²C

Environmental inputs provide the state used to select the next ping.

STM32F446RE · EMBEDDED CONTROL

Firmware in C / C++ on the payload's own MCU

THE NEXT PING IS COMPUTED FROM THE CURRENT ENVIRONMENT



01 ACQUIRE

Read environmental inputs · reject stale / invalid readings

02 DECIDE

Smoothing + hysteresis → CLEAR / TRANSITION / MURKY

03 ADAPT

Centre / bandwidth · pulse duration · amplitude

04 SYNTHESISE

LFM · geometric · Barker-13 × Hann / Hamming / Blackman

05 STREAM

TIM6 + DMA → 12-bit DAC1 / PA4 · 1 MS/s

06 VERIFY

ADC + DMA capture → FFT · spectrogram · matched filter

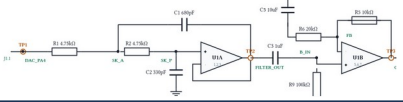
ADAPTIVE NEXT PING

Parameters chosen per water condition

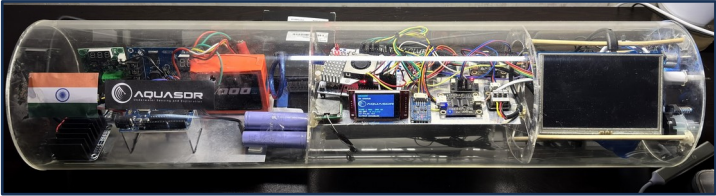
PARAMETER	CLEAR	TRANSITION	MURKY
CENTRE	42 kHz	40 kHz	38 kHz
BANDWIDTH	4 kHz	2 kHz	1 kHz
PULSE	2 ms	4 ms	8 ms
AMPLITUDE	35 %	50 %	62 %

CLEAR → shorter / sharper ping
MURKY → lower band / longer pulse / higher output demand

CONDITION · ANALOG FRONT END



LPF → TLV9062 → output
2nd-order low-pass · 70.7 kHz
op-amp gain ×1.5



AQUASDR PAYLOAD physical acrylic payload prototype for AUV integration

VALIDATION MCU ADC capture · FFT · spectrogram · matched filter
+ external instrument (oscilloscope / spectrum analyzer)

TECHNOLOGIES TO BE USED

Embedded C/C++ · STM32F446RE · ADC · TIM6 · DMA · 12-bit DAC

Analog Sallen-Key LPF · TLV9062

Signal processing LFM · Geometric sweep · Barker-13 · FFT · Windowing

Software React · Node.js · Python · OpenSCAD

